Wyatt Johnston

wyattjohnston@utexas.com | 417.761.4439 | wyatt-johnston.com | github.com/wyattj37

Optimization and stochastic modeling research, applications to energy systems and new technologies — mathematical programming, nonlinear optimization, dynamic programming, and techno-economic analysis in Python

Education

The University of Texas at Austin — Austin, TX
Ph.D. Operations Research and Industrial Engineering — August 2025 – December 2029
M.S. Operations Research and Industrial Engineering — August 2025 – December 2026
GPA: 3.86 / 4.00
Coursework: Machine Learning, Optimization, Stochastic Processes, Game Theoretic Modeling

University of Illinois at Urbana-Champaign — Champaign, IL
B.S. Systems Engineering & Design — August 2021 – May 2025
Concentration + Minor: Computer Science
GPA: 3.98 / 4.00

Research Experience

UNIVERSITY OF TEXAS, Austin, TX — January 2026 - Present

Graduate Research Assistant – Optimization and Energy

- Formulated and solved large-scale mixed-integer programming models in Python for optimized design of CO_2 capture, transport, and storage networks.
- Designed a network generation routine using presolve techniques and graph-theoretic methods to reduce the combinatorial search space prior to optimization.
- Exploited problem structure and applied sequential solving strategies to reduce solve time across large batches of MIP instances.
- Developed an end-to-end pipeline spanning data ingestion, geospatial processing, model construction, solver execution, and output summary.

Graduate Research Assistant – Stochastic Processes and New Technologies

- Analytically characterized a stochastic cost model to derive the optimal deployment strategy for sequential decision-making under uncertainty.
- Solved the corresponding Markov decision process via recursive dynamic programming, conducted policy evaluation and sensitivity analysis.

Honors and Awards

Cockrell School of Engineering Graduate Fellowship — 2025 – Present
Engineering Graduate Teaching Fellow — 2026
Grainger College of Engineering Dean's List — 2021 – 2025
University of Illinois Chancellor's Scholar — 2021 – 2025
Edward S. Fraser Award (Grainger Engineering ISE Department) — 2023
Mildred Maddux and Lisle Abbott Rose Scholarship (Grainger Engineering ISE Department) — 2023
ISE Engineering Visionary Scholarship — 2021 – 2022

Leadership

UNIVERSITY OF TEXAS, Austin, TX August 2026 - Present

President, INFORMS Student Chapter, UT Austin

- Lead the officer team and oversee chapter programming, including student seminars, networking opportunities, and alumni events.

UNIVERSITY OF TEXAS, Austin, TX August 2026 - Present

Mentor, Freshman Introduction to Research in Engineering

- Mentored and instructed a team of freshmen in engineering in their first research project.
- Guided students through formulating a research problem and generating, presenting results.

Teaching Experience

UNIVERSITY OF ILLINOIS, Champaign, IL January 2025 – May 2025

Lab Assistant, Instrumentation and Test Lab

- Led lab sessions, conducting demonstrations and assisting with data collection and analysis.
- Provided detailed feedback on lab reports working to improve students' technical writing.

UNIVERSITY OF ILLINOIS, Champaign, IL August 2022 – May 2025

Lead Engineering Tutor

- Interviewed and trained new tutors, supporting the growth of the tutoring program.
- Offered drop-in tutoring and exam review for a range of core engineering courses.

Work Experience

SIEMENS ENERGY, Orlando, FL June 2024 – August 2024

Intern, Power Diagnostics

- Developed a Python script in Jupyter Notebook to aggregate sensor data from power plants, summarize the performance findings and generate easily digestible tables used in reports.
- Performed time series data analysis to compare the accuracy of thermal models for combustion engines, contributing to improved model precision.

TOTH AND ASSOCIATES, Springfield, MO May 2023 – August 2023

Intern, Planning

- Utilized software to inspect misdiagnosed voltage drops in power distribution systems attributed to temperature settings in model, with findings presented at an industry conference.
- Employed data filtering techniques to identify geographic power usage trends over several years, using these insights to compile a growth plan distributed to clients.

Technical Skills

Languages: Python, R, Julia
Methods: mixed-integer programming, nonlinear optimization, dynamic programming, regression, deep learning, game theory
Optimization: Gurobi, Pyomo, PuLP, AMPL
Computing and Data: SciPy, pandas, PyTorch, TensorFlow, SymPy, scikit-learn
Tools: Jupyter Notebook, Git, LaTeX